Telemetry Peripheral Connector

SM06B-GHS-TB J1

BAS516 D1

BZX585-B5V6.115 D2

NL03KT330 L1

C1 10uF

JP1 RST_EXT

PWR_FLAG

R1 100R RX0

R4 100R TX0

RTS

CTS

MP

+VSW

GND

1uH – DFE252012P-1R0M=P2 L2

TPS631000DRLR U1

C5 22u

R10 511k

R11 91k

3v3I_FB

C6 47u

C7 10u

D5 RED

R14 180R

+3.3V

1uH – DFE252012P-1R0M=P2 L3

TPS631000DRLR U3

C13 22u

R22 806k

R23 91k

5v_FB

C14 47u

C15 10u

D7 RED

R24 180R

+5V

THUNDERFLY G1

U2 ATMEGA1284P-AU

RST_EXT

C2 4u7

RST_PRG

R5 10k

Y1 8 MHz

C3 15p

C4 15p

C8 10u

C9 100nF

C10 100nF

C11 100nF

C12 100nF

PWR_FLAG

RESET

XTAL1

XTAL2

AREF

VCC

AVCC

PA0 A0

PA1 A1

PA2 EXT_DETECTION

PA3 CTS

PA4 RTS

PA5 BTN_USER_A

PA6 BTN_USER_B

PB0 B0

PB1 B1

PB3 CS2

PB4 CS1

PB5 MOSI

PB6 MISO

PB7 SCLK

PC0 SCL

PC1 SDA

PC2 C0

PC3 PC3 TP1

PC4 PC4 TP2

PC5 LED1

PC6 LED2

PC7 LED3

PD0 RX0

PD1 TX0

PD2 RX1

PD3 TX1

PD4 TIMEPULSE

PD5 EXTINT

PD6 GEO_STAT

PD7 C1

J3

MISO

SCLK

RST_PRG

MOSI

SW1 SKRTLAE010

SW2 SKRTLAE010

R2 10k

R3 10k

BTN_USER_A

BTN_USER_B

M1 M2 M3 M4 GND MOUNTING HOLES

FID1 Fiducial

FID2 Fiducial

FID3 Fiducial

Transistors may be removed in future revision,

D3 BLUE

R6 51R

R7 10k

Q1 BC817-25

LED2

D6 GREEN

R15 18R

LED3

D4 RED

R8 82R

R9 10k

Q2 BC817-25

LED1

R12 10k

R13 10k

J2 SM08B-GHS-TB

TIMEPULSE

EXTINT

GEO_STAT

SDA

SCL

TX1

RX1

R17 100R

R18 100R

R19 100R

R20 100R

R21 100R

ThunderFly TFGPS Payload Connector

J4 SM04B-GHS-TB

I²C – Dronecode Autopilot Connector Standard

J5 HEADER_2x15

J6 SM07B-GHS-TB

SPI – Dronecode Autopilot Connector Standard

SCLK

MISO

MOSI

CS1

CS2

R25 100R

R26 10k

R27 100R

R28 100R

R29 100R

R30 100R

EXT_DETECTION

TFUNIPAYLOAD01B

Universal interface for TF-ATMON sensor payload.

Author: ThunderFly s.r.o.

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THUNDERFLY